

LAPORAN UJIAN PRAKTIKUM
REKOGNISI DAN PEMINDAIAN SERVER MENGGUNAKAN NMAP



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1. Tujuan Pengujian

Praktikum ini bertujuan untuk melakukan rekognisi (reconnaissance) dan pemindaian terhadap server menggunakan aplikasi **Nmap**. Pengujian dilakukan untuk memperoleh informasi mengenai port TCP yang terbuka, layanan (service) yang berjalan, versi layanan, sistem operasi yang digunakan, serta mengidentifikasi potensi risiko keamanan pada server target.

2. Metode Yang Digunakan

Metode yang digunakan adalah **Active Reconnaissance** menggunakan aplikasi **Nmap**. Proses pemindaian dilakukan melalui Command Prompt dengan beberapa opsi pemindaian untuk memperoleh informasi mengenai port, service, versi service, sistem operasi, serta jalur jaringan (traceroute).

Tahapan pengujian:

1. Menentukan target server.
2. Menjalankan perintah Nmap.
3. Mengidentifikasi port TCP yang terbuka atau tertutup.
4. Mengidentifikasi service dan versinya.
5. Melakukan deteksi sistem operasi.
6. Melakukan analisis terhadap hasil pemindaian.
7. Menyusun rekomendasi keamanan.

3. Hasil Pemindaian

A. Target 1. www.bgn.go.id

PORT	STATUS	SERVICE	VERSION / KETERANGAN
9/tcp	filtered	discard	-
80/tcp	open	http	CW (CloudWAF)
81/tcp	open	tcpwrapped	-
82/tcp	open	xfer?	-
83/tcp	open	mit-ml-dev?	-
84/tcp	open	ctf?	-
443/tcp	open	https	CW (CloudWAF)

```
C:\Users\asus>nmap -A www.bgn.go.id
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 10:41 +0700
Nmap scan report for www.bgn.go.id (110.239.78.83)
Host is up (0.054s latency).
Other addresses for www.bgn.go.id (not scanned): 110.239.78.90 64:ff9b::6eef:4e5a 64:ff9b::6eef:4e53
rDNS record for 110.239.78.83: ecs-110-239-78-83.compute.hwclouds-dns.com
Not shown: 894 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
9/tcp     filtered discard
80/tcp    open  http   CW
|_http-server-header: CW
|_fingerprint-strings:
|_GetRequest:
|   HTTP/1.1 404 Not Found
|   Date: Mon, 20 Jul 2026 03:42:18 GMT
|   Content-Type: text/html
|   Content-Length: 2178
|   Connection: close
|   ETag: "679257af-882"
|   Server: CW
|   <!DOCTYPE html><html style="height:100%;width:100%"><head><meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
|   <meta http-equiv="Server" content="CloudWAF" /><title id="title">
|   </title></head><body style="height:100%;width:100%;margin:0px;font-family:Microsoft yahei"><div><style type="text/css"> .button
|   {float:right;margin-right:2rem;text-decoration:none;background:white;color:#e94d4c;padding: 5px 15px 5px 15px;font-size:1rem;font-fam
|   ily:
|   |   Arial,Helvetica,Verdana,sans-serif;font-weight:bold;border-radius:0rem;border: none;-webkit-transition:all linear 0.30s;-moz-t
|   ransition:all linear 0.30s;}.button:hover{text-decoration:none;background:#e94d4c;color:#f2f2f2;}</style><input type="button"
|   |
|   |_HTTPOptions:
|   |   HTTP/1.1 404 Not Found
|   |   Date: Mon, 20 Jul 2026 03:42:19 GMT
```

B. Target 2. www.kemenang.go.id

PORT	STATUS	SERVICE	KETERANGAN
80/tcp	closed	http	-
81/tcp	closed	hosts2-ns	-
82/tcp	closed	xfer	-
83/tcp	closed	mit-ml-dev	-
84/tcp	closed	ctf	-
113/tcp	closed	ident	-
443/tcp	closed	https	-
2099/tcp	closed	h2250-annex-g	-
8093/tcp	closed	unknown	-
9000/tcp	closed	cslistener	-
9001/tcp	closed	tor-orport	-
9002/tcp	closed	dynamid	-
9003/tcp	closed	unknown	-

```
C:\Users\asus>nmap -A www.kemenang.go.id
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 10:56 +0700
Nmap scan report for www.kemenang.go.id (103.7.13.243)
Host is up (0.060s latency).
Other addresses for www.kemenang.go.id (not scanned): 64:ff9b::6707:df3
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    closed http
81/tcp    closed hosts2-ns
82/tcp    closed xfer
83/tcp    closed mit-ml-dev
84/tcp    closed ctf
113/tcp   closed ident
443/tcp   closed https
2099/tcp  closed h2250-annex-g
8093/tcp  closed unknown
9000/tcp  closed cslistener
9001/tcp  closed tor-orport
9002/tcp  closed dynamid
9003/tcp  closed unknown
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: firewall|broadband router|general purpose
Running: Cyberoam embedded, Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:2.6 cpe:/o:linux:linux_kernel:2.6.18
OS details: Cyberoam UTM Firewall, Linux 2.6.32, Endian Firewall 2.3 (Linux 2.6), Linux 2.6.18, Linux 2.6.23, Linux 2.6.24
Network Distance: 11 hops

TRACEROUTE (using port 443/tcp)
HOP RTT ADDRESS
1 4.00 ms 192.168.221.243
2 1343.00 ms 10.202.86.2
3 50.00 ms 10.202.86.1
4 56.00 ms 140.213.286.69
5 51.00 ms ae2s0.jktcyb05-cisp01.xl.co.id (112.215.81.42)
6 ...
7 66.00 ms telkomnet.openixp.net (218.100.36.56)
8 68.00 ms 118.97.5.133
9 73.00 ms 118.97.5.134
10 75.00 ms 103.7.13.2
11 52.00 ms 103.7.13.243

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.49 seconds
```

C. Target 3. www.nasa.gov

PORT	STATUS	SERVICE	VERSION / KETERANGAN
80/tcp	open	http	nginx
443/tcp	open	ssl/http	nginx (SSL/TLS aktif, sertifikat valid) WordPress 7.0.2 (HTTP Generator)

```
C:\Users\asus>nmap -A www.nasa.gov
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 10:06 +0700
Nmap scan report for www.nasa.gov (192.0.66.108)
Host is up (0.073s latency).
Other addresses for www.nasa.gov (not scanned): 2a04:fa07:ffff::c000:426c
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    nginx
443/tcp    open  ssl/http nginx
|_http-title: Did not follow redirect to https://www.nasa.gov/
|_ssl-date: TLS randomness does not represent time
|_http-title: NASA
|_ssl-cert: Subject: commonName=nasa.gov
|_Subject Alternative Name: DNS:nasa.gov, DNS:www.nasa.gov
|_Not valid before: 2026-06-13T00:21:09
|_Not valid after: 2026-09-07T00:21:09
|_http-generator: WordPress 7.0.2
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: IPcop 2.0 (Linux 2.6.32) (94%), Linux 3.2 (94%), Linux 4.18 (94%), Linux 5.4 (94%), Linux 2.6.32 (92%), Linux 4.9 (92%), Synology DiskStation Manag
er 7.1 (92%), Linux 3.18 (92%), Tiandy NVR (92%), D-Link DSL-2890AL ADSL router (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 7 hops

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 1.00 ms 192.168.221.243
2 41.00 ms 10.202.86.2
3 41.00 ms 10.202.86.1
4 87.00 ms 140.213.286.61
5 87.00 ms 24203.sgw.equinix.com (27.111.228.36)
6 67.00 ms 2635.sgw.equinix.com (27.111.228.209)
7 64.00 ms 192.0.66.108

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 35.84 seconds
```

4. Analisis Temuan

A. Analisis www.bgn.go.id

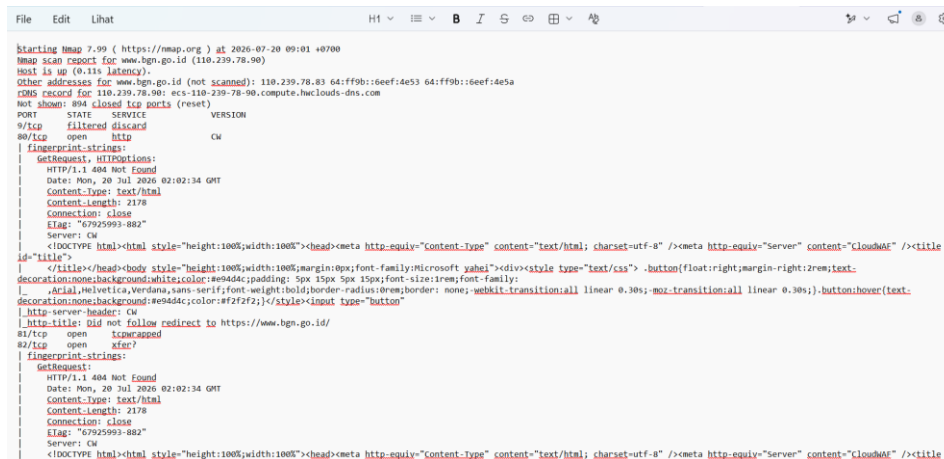
Berdasarkan contoh hasil pemindaian, server menyediakan layanan HTTP dan HTTPS melalui port 80 dan 443. Hal ini menunjukkan bahwa website dapat diakses melalui protokol web standar maupun protokol yang telah menggunakan enkripsi SSL/TLS.

Potensi Risiko

- Port HTTP dapat digunakan tanpa enkripsi apabila tidak dilakukan pengalihan otomatis ke HTTPS.
- Informasi banner server dapat dimanfaatkan oleh penyerang untuk melakukan fingerprinting.

Rekomendasi

- Mengaktifkan redirect otomatis HTTP ke HTTPS.
- Menyembunyikan informasi versi server.
- Memperbarui sistem operasi dan aplikasi web secara berkala.



```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 09:01 +0700
Nmap scan report for www.bgn.go.id (110.239.78.90)
Host is up (0.11s latency).
Other addresses for www.bgn.go.id (not scanned): 110.239.78.83 64:ff9b::6eef:4e53 64:ff9b::6eef:4e5a
RMS record for 110.239.78.90: ecs-110-239-78-90.compute.hclouds-dns.com
Not shown: 894 closed TCP ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
|_ fingerprint-strings:
|_  getrequest, HTTPOptions:
|_  HTTP/1.1 404 Not Found
|_  Date: Mon, 20 Jul 2026 02:02:34 GMT
|_  Content-Type: text/html
|_  Content-Length: 2178
|_  Connection: close
|_  Etag: "67925993-8a2"
|_  Server: CW
|_  <!DOCTYPE html><html style="height:100%;width:100%"><head><meta http-equiv="Content-Type" content="text/html; charset=utf-8" /><meta http-equiv="Server" content="CloudNAE" /><title id="title">
|_  </title></head><body style="height:100%;width:100%;margin:0px;font-family:Microsoft Yahei"><div><style type="text/css"> .button{float:right;margin-right:2rem;text-decoration:none;background:white;color:#e94d4c;padding: 5px 15px 5px 15px;font-size:1rem;font-family:
|_  Arial,Helvetica,Verdana,sans-serif;font-weight:bold;border-radius:0rem;border: none;webkit-transition:all linear 0.30s;moz-transition:all linear 0.30s;}.button:hover{text-decoration:none;background:#e94d4c;color:#f2f2f2};</style><input type="button"
|_  |_  http-server-header: CW
|_  |_  http-title: Did not follow redirect to https://www.bgn.go.id/
81/tcp    open  tcpwrapped
82/tcp    open  xfer?
|_ fingerprint-strings:
|_  getrequest:
|_  HTTP/1.1 404 Not Found
|_  Date: Mon, 20 Jul 2026 02:02:34 GMT
|_  Content-Type: text/html
|_  Content-Length: 2178
|_  Connection: close
|_  Etag: "67925993-8a2"
|_  Server: CW
|_  <!DOCTYPE html><html style="height:100%;width:100%"><head><meta http-equiv="Content-Type" content="text/html; charset=utf-8" /><meta http-equiv="Server" content="CloudNAE" /><title
```

B. Analisis www.kemenag.go.id

Hasil pemindaian menunjukkan bahwa hampir seluruh port berada dalam kondisi **filtered**, sedangkan port 80 dan 443 yang diperiksa berada dalam kondisi **closed**. Selain itu, Nmap mengidentifikasi kemungkinan penggunaan **Cyberoam UTM Firewall** dengan sistem operasi berbasis Linux.

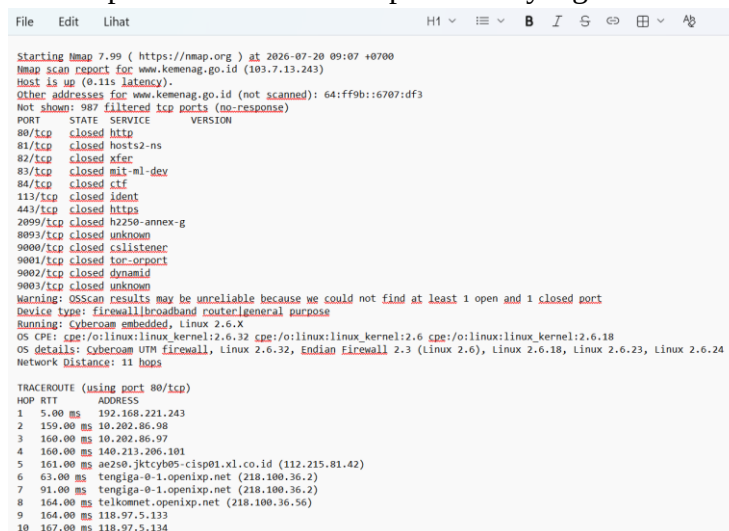
Kondisi tersebut menunjukkan bahwa server memiliki mekanisme keamanan jaringan yang membatasi akses dari luar sehingga proses identifikasi layanan menjadi lebih sulit.

Potensi Risiko

- Informasi sistem operasi masih dapat diperkirakan melalui fingerprinting.
- Administrator tetap perlu melakukan pembaruan firewall dan sistem operasi secara berkala.

Rekomendasi

- Memastikan konfigurasi firewall selalu diperbarui.
- Melakukan audit keamanan secara berkala.
- Menutup seluruh port yang tidak digunakan.



```
File Edit Lihat H1 v B I S G v A
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 09:07 +0700
Nmap scan report for www.kemenag.go.id (103.7.13.243)
Host is up (0.11s latency).
Other addresses for www.kemenag.go.id (not scanned): 64:ff9b::6707:df3
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    closed http
81/tcp    closed hosts2-ns
82/tcp    closed xfer
83/tcp    closed mit-ml-dev
84/tcp    closed ctf
113/tcp   closed ident
443/tcp   closed https
2099/tcp  closed h2250-annex-g
8093/tcp  closed unknown
9000/tcp  closed cslistener
9001/tcp  closed tor-orport
9002/tcp  closed dynamid
9003/tcp  closed unknown
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: firewall|broadband router|general purpose
Running: Cyberoam embedded, Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:2.6 cpe:/o:linux:linux_kernel:2.6.18
OS details: Cyberoam UTM Firewall, Linux 2.6.32, Endian Firewall 2.3 (Linux 2.6), Linux 2.6.18, Linux 2.6.23, Linux 2.6.24
Network distance: 11 hops

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 5.00 ms 192.168.221.243
2 159.00 ms 10.202.86.98
3 160.00 ms 10.202.86.97
4 160.00 ms 140.213.206.101
5 161.00 ms ae250-jktyb05-cisp01.xl.co.id (112.215.81.42)
6 63.00 ms tengiga-0-1.openixp.net (218.100.36.2)
7 91.00 ms tengiga-0-1.openixp.net (218.100.36.2)
8 164.00 ms telkomnet.openixp.net (218.100.36.56)
9 164.00 ms 118.97.5.133
10 167.00 ms 118.97.5.134
```

C. Analisis www.nasa.gov

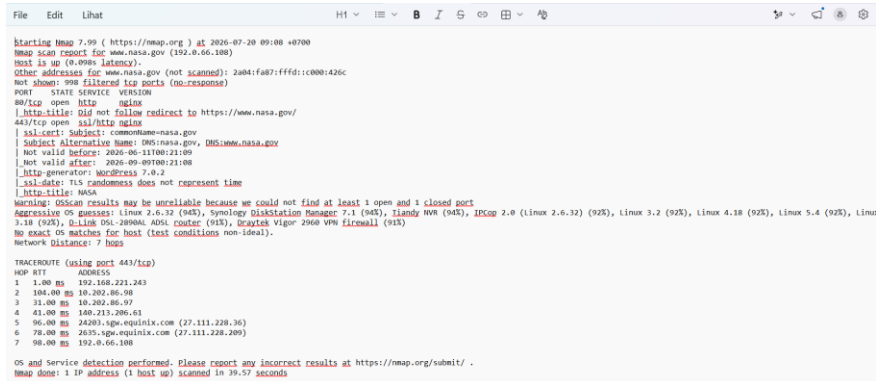
Hasil pemindaian menunjukkan bahwa website NASA menggunakan **nginx** dengan layanan HTTP dan HTTPS yang aktif. Website secara otomatis mengarahkan pengguna menuju koneksi HTTPS dan menggunakan sertifikat SSL yang masih berlaku. Selain itu, Nmap mendeteksi bahwa website dibangun menggunakan WordPress.

Potensi Risiko

- Informasi mengenai teknologi web (WordPress) dapat dimanfaatkan untuk mencari kerentanan apabila tidak diperbarui.
- Sistem operasi tidak dapat dipastikan secara akurat karena keterbatasan kondisi pemindaian.

Rekomendasi

- Selalu memperbarui WordPress beserta plugin yang digunakan.
- Menggunakan konfigurasi TLS terbaru.
- Melakukan monitoring terhadap log server.

A screenshot of a terminal window showing the output of an Nmap scan. The scan was performed on 192.0.66.100 (www.nasa.gov) at 2020-07-20 09:08:40. The output shows that the host is up, and the scan detected an open port 443/tcp. The service running on this port is identified as nginx. The scan also detected an SSL certificate for www.nasa.gov, which is valid and signed by a trusted authority. The scan results are displayed in a structured format with various fields and values. The terminal window has a standard Linux-style interface with a title bar and menu options.

```
Starting Nmap 7.99 ( https://nmap.org ) at 2020-07-20 09:08:40
Nmap scan report for www.nasa.gov (192.0.66.100)
Host is up (0.000s latency).
Other addresses for www.nasa.gov (not scanned): 2a04:fa87:ffff::c000:426c
Not shown: 999 filtered tcp ports (no response)
PORT      STATE SERVICE VERSION
443/tcp    open  http      nginx
|_ http_title: did not follow redirect to https://www.nasa.gov/
443/tcp    open  ssl/http  nginx
|_ Subject: Subjects: commonsense-nasa.gov
|_ Subject Alternative Name: DNS:www.nasa.gov, DNS:www.nasa.gov
|_ Not valid before: 2020-06-11T00:21:09
|_ Not valid after: 2020-09-07T00:21:09
|_ http-generator: WordPress 7.0.2
|_ ssl_date: TLS randomness does not represent time
|_ http_title: none
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 2.6.32 (94%), Synology DiskStation Manager 7.1 (94%), Liandry HVR (94%), IPsec 2.0 (Linux 2.6.32) (92%), Linux 3.2 (92%), Linux 4.18 (92%), Linux 5.4 (92%), Linu
3.10 (92%), Q-Link DSL-2890AL ADSL router (91%), Draytek Vigor 2900 VPN firewall (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 7 hops
TRACEROUTE (using port 443/tcp)
HOP RTT ADDRESS
1 1.00 ms 192.168.223.243
2 104.00 ms 10.202.86.50
3 31.00 ms 10.202.86.97
4 41.00 ms 140.213.206.61
5 96.00 ms 24203.sgw.equinix.com (27.111.228.36)
6 78.00 ms 2639.sgw.equinix.com (27.111.228.289)
7 98.00 ms 192.0.66.100
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 39.57 seconds
```

5. Kesimpulan

Berdasarkan praktikum yang telah dilakukan menggunakan aplikasi Nmap, proses rekognisi berhasil memberikan informasi mengenai kondisi keamanan dari ketiga target yang dipindai.

- www.bgn.go.id (contoh) menunjukkan layanan HTTP dan HTTPS aktif pada server berbasis Linux.
- www.kemenag.go.id menunjukkan sebagian besar port berada dalam kondisi **filtered** atau **closed**, yang mengindikasikan adanya perlindungan firewall sehingga akses dari luar dibatasi.
- www.nasa.gov memiliki layanan HTTP dan HTTPS yang aktif menggunakan **nginx**, sertifikat SSL yang valid, serta website berbasis **WordPress**, sehingga komunikasi data berlangsung melalui koneksi terenkripsi.

Secara keseluruhan, penggunaan Nmap sangat membantu dalam proses identifikasi port, layanan, sistem operasi, serta konfigurasi keamanan server. Informasi yang diperoleh dapat digunakan sebagai dasar evaluasi keamanan dan penyusunan langkah-langkah peningkatan keamanan jaringan.